



LUC ALLAIRE

Software Developer · Kotlin Multiplatform · Android · iOS

Trois-Rivières, Qc, Canada

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[Github](#) · [LinkedIn](#) · [Portfolio](#)

TECHNICAL SKILLS

Mobile	Kotlin Multiplatform, Jetpack Compose, SwiftUI, Ktor, Koin, Room, SKIE, Coroutines/Flow
DevOps & Infra	Docker, TeamCity, GitHub Actions, Ansible, Traefik, Proxmox
Web & Backend Development	Angular, PostgreSQL, Spring Boot, NestJS
Languages	Kotlin, Swift, TypeScript, Go, Python, C#

PROFESSIONAL EXPERIENCE

Simaudio – Software Developer 2025 – Present

- Reverse-engineered a partially documented proprietary TCP protocol to control MiND1 players, delivering a production-ready Kotlin command queue with mutex serialization and a keepalive coroutine.
- Architected a Kotlin DSL TeamCity pipeline backed in Git, leveraged AI tooling to refactor build configurations across 2 apps following a GitFlow rework, and documented the setup thoroughly enough for a mobile developer to independently provision and restore the server.
- Enabled end-to-end Android and iOS deployment to Firebase App Distribution and TestFlight with manual production gating.
- Established unit testing standards in the shared KMP module using the AAA pattern, contributing to continuous integration test coverage across Android and iOS.

UQTR · STI IDEV Team – Software Developer Intern May – August 2024

- Developed a custom Jira project-tracking plugin via Atlassian Forge and refactored the data model of a legacy inventory plugin for stability across the Atlassian suite.
- Engineered a local development bootstrapping CLI in TypeScript, eliminating environment configuration discrepancies and streamlining developer onboarding.

UQTR – Research Assistant Autonomous biopesticide-spraying robot 2023–2024

- Implemented and optimized an OpenCV/Python image-processing pipeline for an autonomous biopesticide-spraying robot, translating visual data into actionable navigation parameters.

SELECTED PROJECTS & ARCHITECTURE

Planific Kotlin Multiplatform · Compose · SwiftUI · SKIE

- Architected a cross-platform app using strict MVI unidirectional data flow — 100% of domain logic, state, and data resides in a shared KMP module.
- Leveraged SKIE for seamless Swift interop, exposing Kotlin StateFlows as native Swift AsyncSequences for the iOS UI.

Home Ops Infrastructure Docker · Ansible · Traefik · Debian

- Provisioned a 3-node bare-metal Debian cluster managed via 12 custom Ansible roles, featuring a dual-path network mesh with NetBird and Cloudflare tunnels.
- Deployed a Traefik sidecar with automated Uptime Kuma monitoring and Authelia SSO integration.

ASUQTR — Autonomous Submarine Linux · ROS · 2023–2024

- Migrated the team's development server from Windows 10 with Atlassian tools to Ubuntu with open source alternatives.
- Designed and integrated the computer vision system (OpenCV/Python) for autonomous navigation in the RoboSub competition.

EDUCATION

B.Sc. Computer Science *Dean's Honor List 2023 & 2024* Université du Québec à Trois-Rivières · 2022 – 2026

B.Sc. Chemistry Université du Québec à Trois-Rivières · 2020 – 2022